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Optimization of Paid Search Traffic Effectiveness and Users Engagement within Websites

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Abstract: Optimized paid search advertising campaigns composed of multiple data analytics insights and prior experiences of search engine marketing performances. However, when advertisers compete in the battle of paid search ads rankings in search engines, complexity in optimization is increased. The higher the ranking position of search ads are, the higher the probability to be clicked by the search engines users. Despite the existing knowledge of the factors that contribute to the higher ranking position in search ads, such as proper relevancy amongst users' search terms, text ads and landing pages content, little is known about search engines users' behavior after ads clicking. Low interaction or immediate abandonments from the landing pages potentially leads to a waste of budget spent on each paid advertising campaign. In this respect, advertisers should pay much more attention to the engagement of paid traffic visitors after clicking on search ads, and not only on ads rankings and rates of search impression shares in search engines. In this paper, the authors develop a computational data-driven methodology with a purpose to estimate and predict paid traffic visitors' engagement in seven courseware websites after clicking on the search ads. The higher the engagement with the landing page is, the higher will be the probability for conversions. At the first stage, web behavioral analytics are retrieved for 120 consecutive days in certain web metrics such as the volume of paid traffic visitors, the average pages per session, the average session duration and the bounce rate. Statistical analysis of the extracted web behavioral datasets takes place for understanding the cohesion, validity, and intercorrelations between the web metrics. KMO and Bartlett's test of Sphericity and Pearson coefficient of correlation are adopted. It is noted that, in an overall point of view, results indicate low rates of engagement after search ads clicking. One step further, agent-based modeling and simulation is adopted as a methodology for abstracting and calibrating paid traffic visitors' behavior inside the examined websites. Poisson distributions are implemented for predicting the potential engagement of paid traffic visitors in specific date ranges. Through this, the paper highlights its practical contribution to decision makers with the purpose to develop search engine marketing campaigns composed of relevant to the users search ads and sufficient content engagement after ads clicking.

Keywords: Digital Marketing, Search Engine Marketing, Search Advertising Optimization, Web Analytics, Behavioral Analytics, Data-Driven Modeling, Agent-Based Models in Marketing, Modeling and Simulation in Marketing

The Context of Paid Search Advertising

Paid search advertising constitutes one of the most well-known methods in the context of strategic digital marketing. It is revolved around three different angular

stakeholders, the creators of the ads (*advertisers*), the receivers of the ads (*search engines' users*) and the search engines as well (Aslam and Karjaluo, 2017). From the advertisers' point of view, the development of effective search ads that express a high click-through rate, lower cost-per-click, and sufficient conversion rate, is a complex and multidimensional task. It requires prior data insights about users' search behaviors and the search terms they use, benchmarking and retrieval of competitors' analytics regarding the average cost per click for bid estimations, plus also the required knowledge to manipulate the dashboards for ads creation such as Google or Bing Ads platforms.

In simplicity, paid search advertising works through a path-dependence process of three sequential stages. In the first stage, users type search terms in search engines and then, in the search engines' results pages (SERPs) two types of results are listed, organic and paid. The paid search ads differentiate their appearance from the organic search results with a notable indication of *Ad.*, with ad extensions such as sitelinks, callouts, structured snippets, star ratings and so on. In the second stage, users click on the webpages that are listed in paid results. The likelihood to click on a specific ad, related to the relevancy between users' search terms and the text of the ad (Google Ads Help, 2019a), the eye-catching structure of the ad with extensions (Google Ads Help, 2019b) and the ranking that the ad receives in SERPs (Ghose and Yang, 2009). In the third stage, after clicking on the ad, users enter the website, navigate, interact and proceed to the possible coverage of their informational or commercial needs with the content of the landing webpage. Otherwise, they abandon the website and jump to another.

For Google -as one of the leaders in search ad market share-, the process of ranking search ads in SERPs is related at least to five core factors that are well-known and comprehensible to advertisers. These are:

- i) The relevancy of users search terms with the terms that the text ads have
- ii) The structure of ads based on using all the available options such as the extensions
- iii) The relevancy of the terms that the search ad have with the landing page text content
- iv) The ease-of-navigation and the mobile-friendliness of the landing page after clicking the ads
- v) The budget that the advertiser is willing to afford.

All the above factors work drastically to influence the Google ranking algorithm for listing the paid search ads (Google Ads Help, 2019c), while at the same time it is observable that the one who pays higher, however, has not a good quality of ads, will not win in the "*bid-battle*" of paid ads listing. Although the ranking algorithm follows a user-centered approach to list the paid search results, a research gap is raised regarding the engagement between users and the content of the landing page, after they click on the search ad from the SERPs. What happens after users click on the ads? How much they interact with the content of the webpage that they land after the search ad click? Does the level of engagement is sufficient to increase the probability for more conversions? Or controversially the

content usage is poor, leading in this way advertisers to spend their money without satisfying the informational or purchasing needs of paid search ads visitors?

In this paper, the authors develop a data-driven methodology for evaluating the engagement between users and websites content after they click search ads in SERPs. Dive deep into web behavioral analytics, crucial metrics are examined such as the average session duration, pages per visit and percentage of immediate abandonments with the purpose to understand paid traffic visitors' engagement with the content of the websites. One step further, the authors propose a predictive model through the *agent-based* computational process. The proposed model, estimates in specific time ranges, the optimal engagement between users and the websites' content they receive in order to cover their informational or consumer needs.

Related Background

According to web behavioral stats for search engines, the Google search engine receives ~63.000 submissions of search queries per second (Seotribunal.com, 2019). This kind of information is able to motivate advertisers to optimize their overall search engine marketing (*SEM*) strategy. More specifically, search terms submissions to search engines' can be set under the generic context of big data. Search terms produce large volumes of search queries in search engines, through high velocity (~63.000 queries per second), while they can be generated through a variety of devices such as desktops, tablets or smartphones. Advertisers are able to exploit these massive datasets regarding users' search behavior, with the purpose to find hidden gems for their paid advertising optimization. They are able to receive insights about popular search terms that could be exploited to the potential development of search ads campaigns for achieving higher click-through-rate and lower cost-per-click.

However, as cited in Aslam and Karjaluo (2019) even if Google clearly express some quality factors that influence search ads ranking in SERPs, prior studies indicate that when click-through-rate and conversion rate increased, cost of ads is increased as well, even if websites move up and down to the ranking hierarchy (Narayanan and Kalyanam, 2015; Nabout and Skiera, 2012). A fact that is also verified by Google through the replacement *Avg. Position* of the ad in the SERPs web metric with the *top* and *absolute top impressions* metrics (Google Ads Help, 2019d). To this respect, it should be noted that advertisers have to pay less attention to the ranking position of their ads, and much more attention the way users interacting with the content of the websites after clicking on the search ads. That is, understanding how paid traffic visitors interact within the content of the landing pages, constitutes a decisive step for optimizing probabilities for more conversions, rather than collecting ad rankings data for forecasting potential ads clicking behavior.

In this regard, fully exploiting the capabilities that search engine ads platforms give for developing attractive ads, is an essential advertising part of the whole

SEM strategy. Nevertheless, it does not solely ensure conversions without having landing pages that users express a sufficient engagement rate on them. Against this backdrop, in this paper, the authors proceed into a web mining process of web behavioral analytics regarding paid traffic visitors' engagement in seven courseware websites for 120 consecutive days. Paid traffic visitors engagement data metrics are retrieved regarding the average session duration, the pageviews per visit and the percentage of immediate abandonments with no interaction (*bounce rate*). Through the retrieval of these metrics, Pearson's ρ of coefficient correlation is implemented with the purpose to examine the probability of positive or negative intercorrelations between the metrics and thus paid traffic visitors level of engagement after clicking the search ads.

Positive correlations between web analytics metrics indicate that paid traffic visitors express an analogous engagement with the content in terms of the number of pages they browsing and the time they spend after clicking the search ads. Controversially, negative intercorrelations indicate that paid traffic visitors express low rates of engagement as they might visit the websites, however, they do not browse in many webpages and they spend less time inside the websites. To this respect, three research hypotheses are presented:

H1: The amount of Total Paid Traffic Visitors, express a positive correlation with the Number of Pages that they browsing.

H2: The amount Total Paid Traffic Visitors, express a positive correlation with the Average Session Duration.

H3: The amount of Total Paid Traffic Visitors, express a negative correlation with the percentage of the Bounce Rate.

Furthermore, a data-driven computational development of an agent-based model takes place. This is based exclusively on the statistical results of the examined web behavioral analytics datasets. Both statistical analysis and agent-based modeling and simulation will give the advantage to advertisers to:

- a) Understand the satisfaction of paid search visitors in terms of the content they receive
- b) Predict the highest and lowest engagement rates in potential specific date ranges
- c) Adopt or prevent the content strategy that was implemented and resulted in high/low levels of paid traffic visitors' engagement through the specific date range of 120 days.

In the next chapter, the research methodology of the paper is presented in a more comprehensive way.

Methodology

The purpose of this study is composed of two main pillars. Firstly, to establish a framework for measuring and evaluating paid traffic visitors' engagement in websites after clicking the search ads. Secondly, to develop a computational model that predicts the lowest and highest paid traffic visitors' engagement with the content in specific date ranges. This will be an advantage for advertisers to make

comprehensible what kind of content of the website, receives higher paid traffic visitors' engagement with the purpose to adopt it to potential content development.

Measure and evaluation of engagement: For the measurement and evaluation of paid traffic visitors' engagement, the authors proceeded into the retrieval of four web behavioral analytics metrics, the average pages per session, the average session duration, the bounce rate and the total number of paid search traffic visitors. Aiming to improve the decision makers' data analytics competency (Ghasemaghaei, Ebrahimi, and Hassanein, 2018) and more specifically to increase the level of web analytics exploitation (Järvinen and Karjaluoto, 2015), the authors explain each one of the selected metrics.

- **Total Paid Search Traffic Visitors:** Indicates the total number of users that exclusively visit the websites after clicking on the search ads of SERPs.
- **Average Pages per Session:** Assume that a user enters the website for a specific amount of time and visits x number of pages to the total y number of pages that the whole website contains. Hence, the average pages per session estimates the average number of pages that the total number of users explore within the time they spend after visiting the website.
- **Average Session Duration:** It is related to the average time in minutes that users spend inside the website. Within the duration of the session, several activities might be accomplished such as multiple pages browsing, forms submissions, purchases or any other kind of activity that can be determined as conversion.
- **Bounce Rate:** This web metric indicates the percentage of users that visit the website, however, they leave immediately with no any kind of interaction with the content. This entails that users might enter the website, however, they immediately abandon the content and then return back to the SERPs.

For the collection of the four metrics of the examined seven courseware websites, the SEMRush and the SimilarWeb web behavioral analytics platforms were used. The four examined web analytics metrics were retrieved for 120 sequential days. Within this specific date range, multiple fluctuations were observed in the numerical values of metrics. This fact is indicated in Table 1., while observing several variations in descriptive statistics of min, max, median and standard deviations for each one of the web metrics.

Predictive Model Development: For the development of a predictive model that computes the potential highest and lowest users' engagement rates, the authors proceed into the adoption of the agent-based modeling (ABM) method. With ABM it is possible to represent at an abstraction level the *behavior* of *agents*. The behavior of agents depends on multiple computational *conditions* of if-then rules that are *nonlinear* (Bonabeau, 2002). The possible behavior agents could be examined in a free-risk environment of multiple experiments under the context of simulation runs. At the same time, ABM gives the advantage to decision makers for describing the *system's* features and *forecasting* future scenarios about an opti-

mized behavior of agents within the system (Barbati, Bruno, and Genovese, 2012). In this case of examining paid traffic visitors engagement after clicking search ads:

- The agents are the paid traffic visitors
- The systems are the examined courseware websites
- The behavior related to the engagement that paid traffic visitors express regarding the Average Pages per Session, the Average Session Duration and the Bounce Rate
- The conditions composed by the statistical correlations of the aforementioned web metrics
- The nonlinearity is expressed by the different interactions of every paid traffic visitor (the agent) with the content after clicking the search ad
- The forecasting process is implemented by the multiple, repetitive and simulation runs of the model with the purpose to indicate the optimal paid traffic visitors' engagement within the content in specific date ranges.

Through the duality of methodological approach (*a)Measure and evaluation of engagement - b)Predictive Model Development*) we try to shift from the coverage of web behavioral analytics explanation to the growth and optimization of paid traffic visitors engagement inside the examined courseware websites. On the one hand, rich behavioral data analytics are harvested in certain web metrics. On the other hand, ABM development is composed through a data-driven way, preventing a lack of behavioral data availability (Kennedy, 2012) and most importantly, depicting in this way the realism that a predictive model needs to (Kavak et al. 2018; Bruch and Atwell, 2015). In the next chapter, the statistical results are presented. In addition, ABM development and description take place.

Research Results

In Table 1., the results of the examined possible correlations take place. Controversially of what was expected through the aforementioned hypotheses, the analysis of ρ coefficients expresses negative correlations between the metrics. Regarding the first one hypothesis, the average pages per session are decreased, as more and more paid traffic visitors are coming to the website, indicating in this way a negative correlation with $\rho d(118) = -,332$ and $sig.=,000$ two-tailed. That is, the more visitors are coming to the examined courseware websites through paid search advertising, the fewer pages are navigating per session. Thus, in the first hypothesis, there is a negative correlation between total paid search traffic visitors and average pages per session. In the same line, the *average session duration* is decreased as the volume of paid traffic visitors is increased. This indicates a negative correlation with $\rho d(118)= -,415$ and $sig.=,000$ two-tailed, producing in this way contrasting results with the second one research hypothesis.

In terms of the level of engagement between paid traffic visitors and courseware websites' content, both the average pages per session and the average session duration depicting low levels of content usage. The results can be

strengthened while taking into consideration the third hypothesis that examines the possible negative correlations between the percentage of bounce rate and the volume of paid traffic visitors.

Table 1. ρ Coefficients between the examined web analytics metrics

Web Analytics Metrics	Total Paid Traffic	Pages per Session	Average Session Duration	% Bounce Rate
Total Paid Search Traffic Visitors	1			
Average Pages per Session	-,332	1		
Average Session Duration	-,415	,648	1	
% Bounce Rate	,212	-,711	-,612	1

Total Paid Traffic Visitors: Mean: 67,357 | Mode: 38,700 | Std. Deviation: 2,540 | Min: 18,231 | Max: 250,000

Average Session Duration: Mean: 5,42 | Mode: 4,36 | Std. Deviation: 1,51 | Min: 2,32 | Max: 7,29

Average Pages per Session: Mean: 3,79 | Mode: 3,21 | Std. Deviation: 1,12 | Min: 1,94 | Max: 6,17

% Bounce Rate: Mean: 41,17 | Mode: 40,03 | Std. Deviation: 6,42 | Min: 31,57 | Max: 54,46
KMO and Bartlett's test of Sphericity: ,736 with sig. ,000 | % Variance Explained: 74,169
N=120 days

More specifically, the Pearson coefficient indicated a positive correlation with a with ρ $d^f(118) = ,212$ and $sig.=,000$ two-tailed. That is, the percentage of the bounce rate is increased as the volume of the paid traffic visitors is increased. This might entail that although search engines users click on the search ads, the leave immediately from the landing pages without any kind of interaction with the content. A fact that is related to several reasons such as the irrelevance between users' search terms in the search engine with the text that the landing pages contain, or even with technical issues such as low level of mobile adaptiveness or low loading speed of the examined courseware landing pages. With the purpose to understand the highest and lowest levels of paid traffic visitors' engagement with the content of the examined courseware websites, the authors proceed into the development of the agent-based model. This will work as a very useful decision making supportive mechanism in order to understand the satisfaction of paid search visitors as regards the content they receive after clicking the ads. And hence, adopting or preventing in this way potential high or low levels of engagement.

Agent-Based Model Development

In order to develop the ABM, the authors relied on the adoption of the AnyLogic 8.4 simulation engine. In the next figure (Fig.1), the model development takes place at an abstraction level, describing in this way paid traffic visitors' engagement inside the examined websites. Thought a path-dependence process, the paid traffic visitors (the agents) moving from the one statechart (square shapes) to the others.

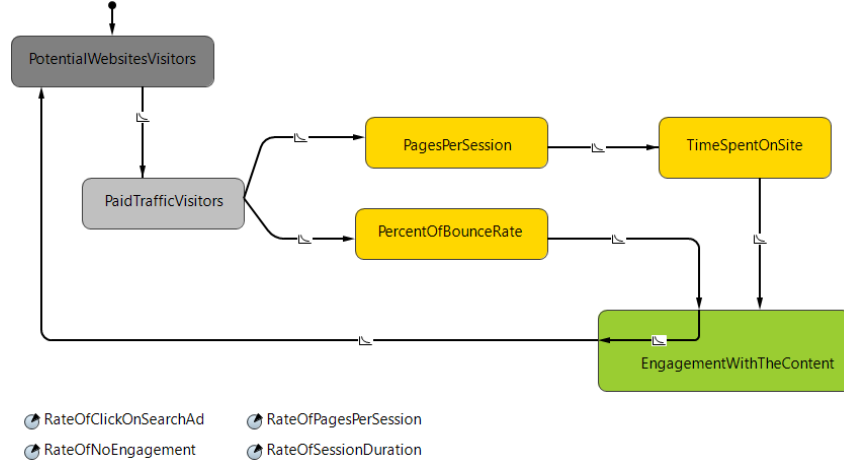


Fig 1. Agent-Based Model Development and Abstraction of Paid Traffic Visitors Engagement inside Courseware Websites

The transitions of the agents between the statecharts (*arrows* in fig.1) are computed via Poisson distributions and their characteristics. More specifically we proceed to the selection of Poisson distribution while:

- i. There is a standard time range of paid traffic visitors' interaction (*120 days of engagement*).
- ii. It has a known constant mean rate of values of engagement such as the *average pages per session*, the *average session duration* and the percentage of *bounce rate*.
- iii. Each one paid traffic user interaction is independent and does not affected by the previous ones.

In addition, it needs to be mentioned that the knowledge of the constant mean rates coming from the preliminary descriptive statistics results in order to set valid lambda λ values in each one condition that the model contains. The preliminary descriptive statistics results derived after the analysis of the retrieved web behavioral analytics for 120 sequential days. The conditions that trigger and compute the transitions of agents between the statecharts are the *RateOfClickOnSearchAd*, the *RateOfNoEngagement*, the *RateOfPagesPerSession*, and the *RateOfSessionDuration*. This leads to foster in a better way a data-driven ABM development (Kavak et. al. 2018) for estimating potential paid traffic visitors' interactions. At the first statechart, the *PotentialWebsiteVisitors* are depicted, following in this

way basic principles of ABM development and the initial agents' state (An, 2012; Fontaine and Rounsevell, 2009).

The *PotentialWebsiteVisitors* indicate the potential action of entering the website through paid search advertising, and becoming in this way *PaidTrafficVisitors*. Following the if-then rule mindset of ABM, there are two processes that paid traffic visitors should follow after entering the website. *a)* Proceed into a sufficient amount of interaction through *PagesPerVisit* and *TimeSpentOnSite* interacting in this way with the content after clicking the search ad. *b)* Proceed into no kind of interaction and abandon the website, increasing in this way the *PercentOfBounceRate*. Even if paid traffic visitors proceed into the immediate abandonments and leaving the websites, they contribute negatively to the final statechart of paid traffic visitors, namely the *EngagementWithTheContent*. Through this way, the paid traffic visitors that proceeded into immediate abandonments after clicking the search ads returned once more to the *PotentialWebsiteVisitors* statechart. Controversially, as paid traffic visitors browsing on several webpages inside websites and spend time inside, they contribute positively to the *EngagementWithTheContent* statechart.

In figure 2, (fig 2.) we present the graphical distribution of two main entities of paid traffic visitors, those who engaged with the content and those who proceeded into immediate abandonments increasing in this way the Bounce Rate percentage. For the graphical visualization of agents' distributions, the authors adopted the mode statistical values of Average Pages per Session and the Average Session Duration. These values are both computed with the purpose to set a percentage of overall content engagement at the *EngagementWithTheContent* statechart. In addition, the minimum value of the Bounce Rate is included in order to encapsulate specific time ranges where low number of paid traffic visitors proceed into immediate abandonments. In this way, the multiple simulation runs of the ABM gives the advantage to estimate the potential engagement of paid traffic visitors after clicking the search ads in a certain period of time of 120 days.

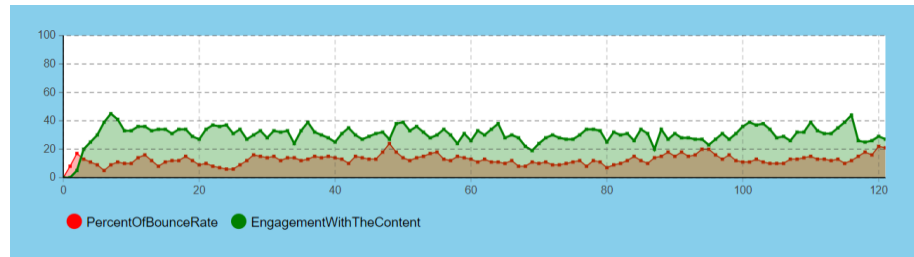


Fig 2. Graphical representation of visitors' engagement with the content of the websites based on the examined time range of 125 days.

In the next table (Table 2.) the highest percentages of engagement with the content and the lowest percentages of bounce rate are presented.

Table 2. Specific date ranges that resulted in the highest engagement with the content and the lowest percentages of Bounce Rate.

EngagementWithTheContent			
Specific Days (of 120 total)	Highest % of EngagementWithTheContent	Specific Days (of 120 total)	Lowest % of Bounce Rate
6	39	20	9
7	41	21	10
8	45	22	8
110	39	23	7
115	39	24	6
116	44	25	6

Through this, the decision makers will be able to understand specific date ranges that resulted higher rates of content usage when potential paid traffic visitors clicking on search ads. For example, there are high percentages of engagement with the content between the 6th, 7th and 8th day with 39%, 41%, and 45% respectively. In the same line, there are very low percentages of Bounce Rate between the 20th and the 25th day with rates ranged from 6% to 10%.

Discussion and Potential Research Steps

In this paper, the authors tried to develop a computational data-driven methodology with a purpose to estimate and predict paid traffic visitors' engagement in seven courseware websites after clicking on the search ads. The higher the engagement is, the higher will be the probability of conversions such as registrations, leads or even purchases. In the next lines, the practical contribution of the paper is presented in a detailed manner.

Contribution to web analytics competency: the era of web analytics exploitation for understanding users' behavior in search engines and websites, promises optimization to personalized content dissemination and coverage of websites visitors' informational or purchasing needs. However, converting web behavioral analytics into useful insights for potential marketing optimizations is not an easy-starting task. Setting key point indicators (Saura, Palos-Sánchez and Cerdá Suárez, 2017; Chaffey and Patron, 2012) and defining in an explicit manner the web analytics metrics, is one of the fundamental steps for exploiting behavioral datasets in order to optimize content usage and dissemination (Järvinen and Karjaluoto, 2015; Pauwels et al. 2009). In this paper, the authors set up as a key point indicator the estimation of paid traffic users' engagement after clicking the search ads in search

engines. Furthermore, the examined web behavioral metrics defined in an explicit manner with the purpose to understand the engagement of the users inside the courseware websites. This also leads to the potential comprehension of possible intercorrelations between the web metrics (Lipe and Salterio, 2002) and which of them are critical to be implemented to the evaluation of online users' behavior in web systems (Drivas, Sakas, and Reklitis, 2019; Weischedel and Huizingh, 2006; Welling and White, 2006).

Practical evaluation tool: In this respect, the proposed methodology of the paper, constitutes a practical evaluation tool for understating if the paid traffic costs of organizations respond effectively to the proper advertising of services to the paid traffic visitors. For example, if the organizations spend a high budget on paid search advertising, however, the paid traffic visitors act with low rates of engagement or a high percent of bounce rate, then the unnecessary waste of economic resources is just around the corner. Thus, practical quantitative measurement is needed, in order to evaluate paid search traffic effectiveness and then to optimize the text and the layout of websites content for higher engagement with visitors. In this paper, the statistical results indicated low levels of engagement after users clicked search ads. It is very important to mention that all hypotheses resulted in negative values of engagement. More specifically, the average pages per session and the average session duration are decreased as more and more paid traffic visitors are coming to the examined courseware websites (H1 and H2). On the other side, the bounce rate is increased as the volume of paid traffic visitors is increased as well (H3).

Therefore, there are indications that the performance of paid search traffic campaigns are quite low in the examined courseware websites. And even if the authors adopted the agent-based method for predicting potential optimized paid traffic engagement via Poisson distributions, rates of engagement did not exceed the 44% or 45%. This works as a corroborative indication that paid search traffic engagement after search clicks still remained low after simulation model runs. Furthermore, in terms of paid search advertising ranking algorithms (Google, 2019a; 2019b; 2019c; 2019d), it is very crucial to proceed into further research approaches in order to completely understand if Google also takes into consideration the paid traffic users engagement for ranking adjustment in SERPs. This will give an additional advantage to advertisers for creating in a strategic marketing manner the content of the landing pages, ensuring in this way an increased to the time users spend and the pages they browsing after ads clicking.

Data-driven agent-based modeling contribution: From the initial steps of ABM as a computational abstracting methodology for describing the complexity of agents within a system till nowadays, it is criticized that model development is more an art, rather than science (Grigoryev, 2015; Grignard et al., 2013; Bonabeau, 2002). However, it is noted that the exploitation of web behavioral datasets works as a new milestone to improve ABM adoption for problem description and optimization. For example, in this paper, there was prior experience about paid traffic visitors' possible actions inside the websites. This worked drastically to de-

velop the model at a specific abstraction level. But the computational transitions of agents within the states derived exclusively from the analysis of web behavioral datasets. A fact that prevents the lack of available behavioral data, suitable for abstraction levels (Kavak et al. 2018; Bruch and Atwell, 2015). This also leads to well-informed ABM developments that relied both on experience and data-driven indications. In regard to the above-mentioned statements, the proposed model in this paper is able to describe, evaluate and predict paid traffic visitors engagement as agents. Nevertheless, further research is needed to examine the model accuracy and flexibility if more conditions and entities will be included that multiply and complex paid traffic visitors' behavior that govern the examined courseware websites (Klabunde and Willekens, 2016). Furthermore, it should not be omitted the possible hybridity that predictive models for marketing purposes include. This perspective combines at the same time agent-based and systems dynamics approaches for estimating optimal dissemination of organizational resources (Drivas et al. 2017; Sarlis, Drivas and Varveris, 2017). This further research contribution is *not-deterrent* as the web analytics topic expands with new emerging web performance metrics, as many of them involve the allocation of financial resources.

Concluding this research approach, it should be noted that this study constitutes a part of a broader effort to establish a methodological framework of exploiting web behavioral analytics, regardless of the organizational sector and the volume of observed and harvested behavioral datasets. Further research is needed in order to prove that paid traffic visitors present similar or different behaviors after clicking search ads in different sectors rather than e-learning environments such as e-commerce. This will be a decisive step to open new quantitative perspectives for the informational and purchasing needs of potential customers and how they behave within the web systems and their content.

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